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Exploring the vast landscape of stable materials for CO_2RR

A resource-efficient computational screen for electrochemical CO_2 reduction catalysts

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150,000 → 10

Copper as the benchmark catalyst

29

Cu

state of the art

- The only metal that binds CO* while suppressing the competing hydrogen evolution reaction
- Reduces CO_2 to valuable multi-carbon (C_{2+}) products
- Yet neither active nor selective enough to compete with fossil-based fuels

MOTIVATION

Reported activity of alternative catalysts

- Total CO_2RR current from 11 studies, normalised to electrochemically active surface area
- Few catalysts differ significantly from polycrystalline Cu
- Apparent gains (CuAg, CuNi, GaNi) are as sensitive to the reported ECSA as to intrinsic activity

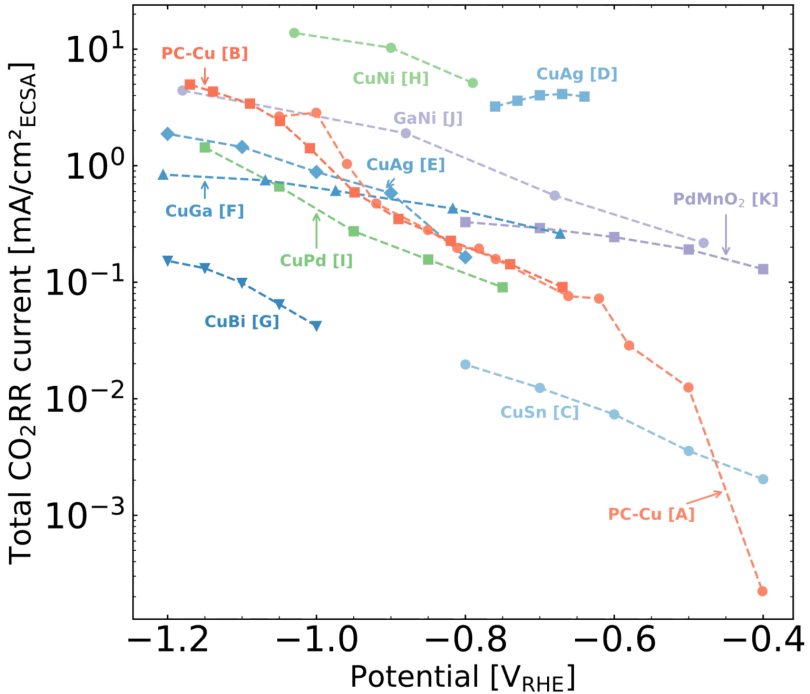


Fig. 1 · Literature review, total CO_2RR current vs. V_{RHE}

APPROACH

Three screening criteria

01

Stability

Must survive CO_2RR conditions: pH 6–8, −1 to −2 V_{SHE}

02

Binding energies

Moderate CO* and weak H* binding — the region copper occupies

03

Work function

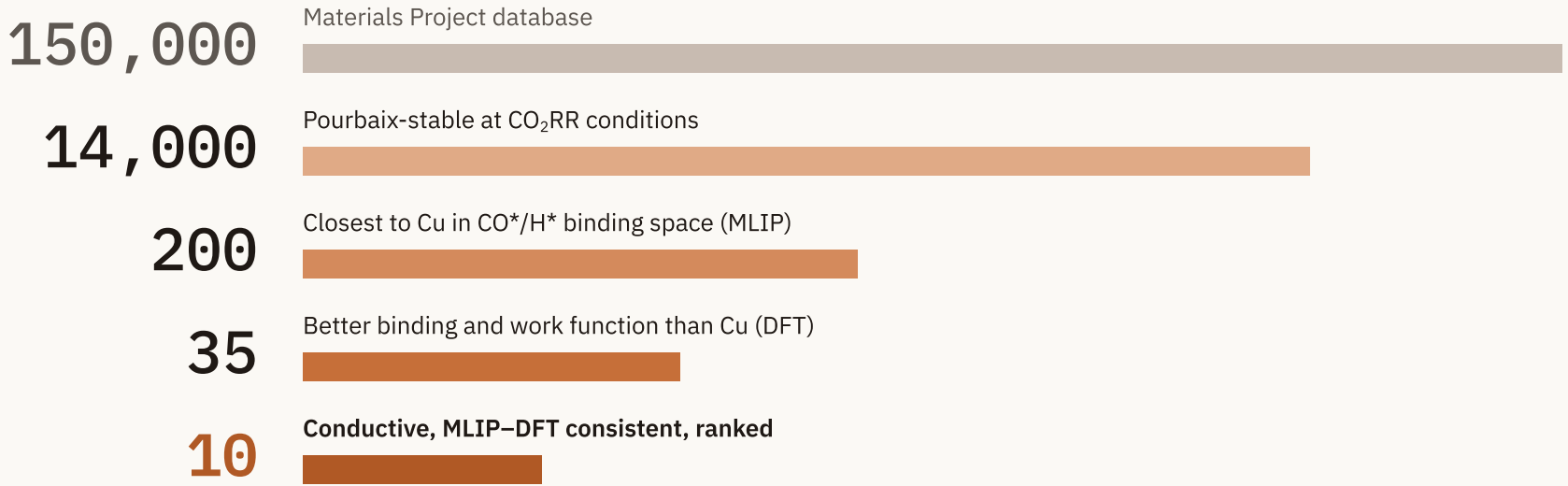
A low work function correlates with suppressed hydrogen evolution

Pourbaix diagrams · Materials Project Pre-trained MLIP · DFT single points DFT · GPAW, RPBE

Karlsson et al. · Electrochim. Acta 549 (2026) 148053

APPROACH

Screening workflow



Karlsson et al. · Electrochim. Acta 549 (2026) 148053 · bars on log scale

FILTER 1 OF 3

Pourbaix stability

- Pourbaix diagram for every Materials Project entry (pymatgen)
- Stable if decomposition energy < 0.2 eV/atom
- Window: pH 6–8 and −1 to −2 V_{SHE}

~11×

fewer candidates, with no new DFT calculations

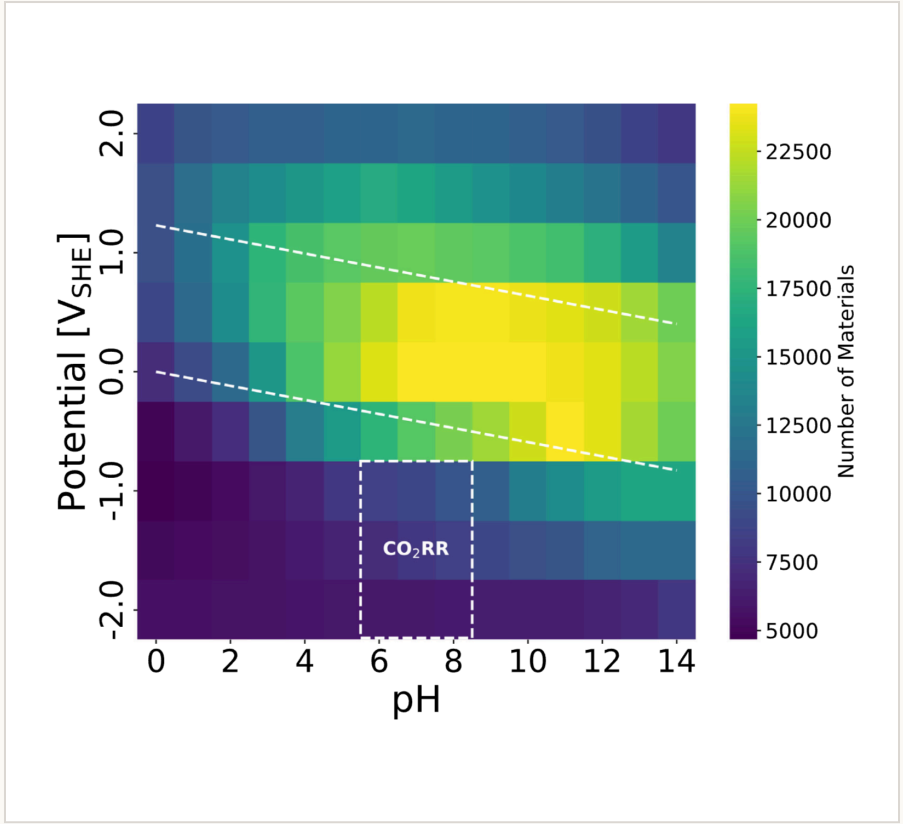


Fig. 2 · Stable materials per pH/potential; box = CO_2RR window

FILTER 2 OF 3

ML-estimated binding energies

- Pre-trained EquiformerV2 MLIP (OC20), no fine-tuning
- One slab per material; Miller index set by space group
- Metals reproduce the trends of our earlier DFT study

200 materials closest to Cu passed on to DFT

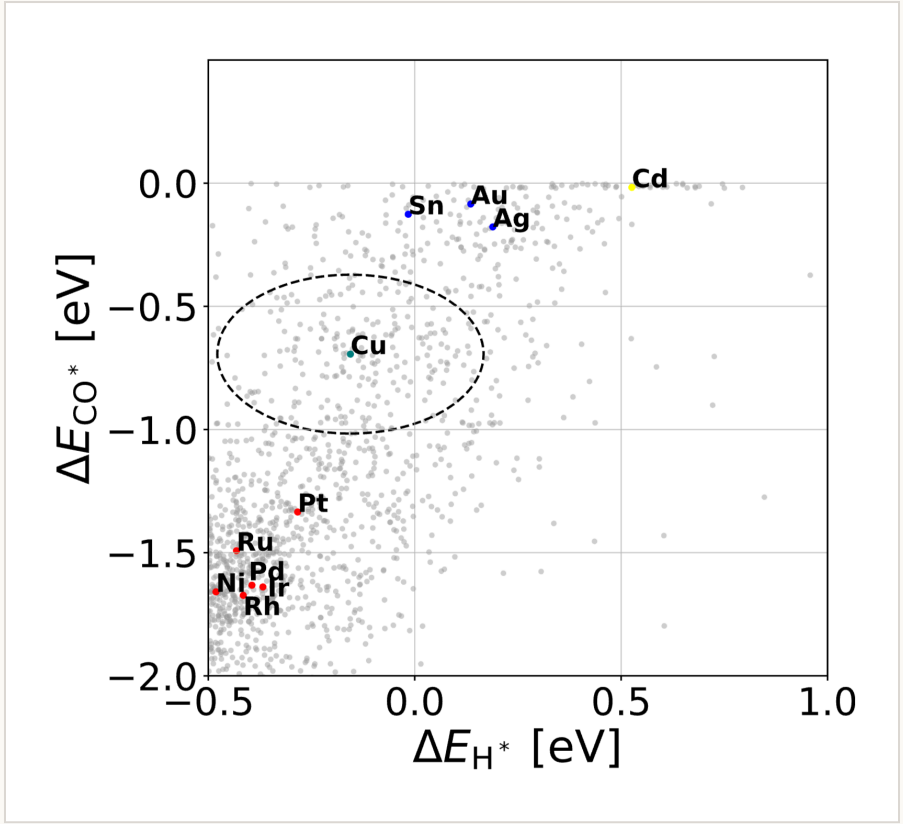


Fig. 3 · MLIP ΔE_{CO*} vs. ΔE_{H*}, ~14,000 materials

FILTER 3 OF 3

DFT validation and work function

- RPBE single points on MLIP geometries, no relaxation
- Keep the quadrant below-right of Cu: stronger CO*, weaker H*
- Work function below Cu (4.67 eV)

35 materials beat Cu on all three descriptors

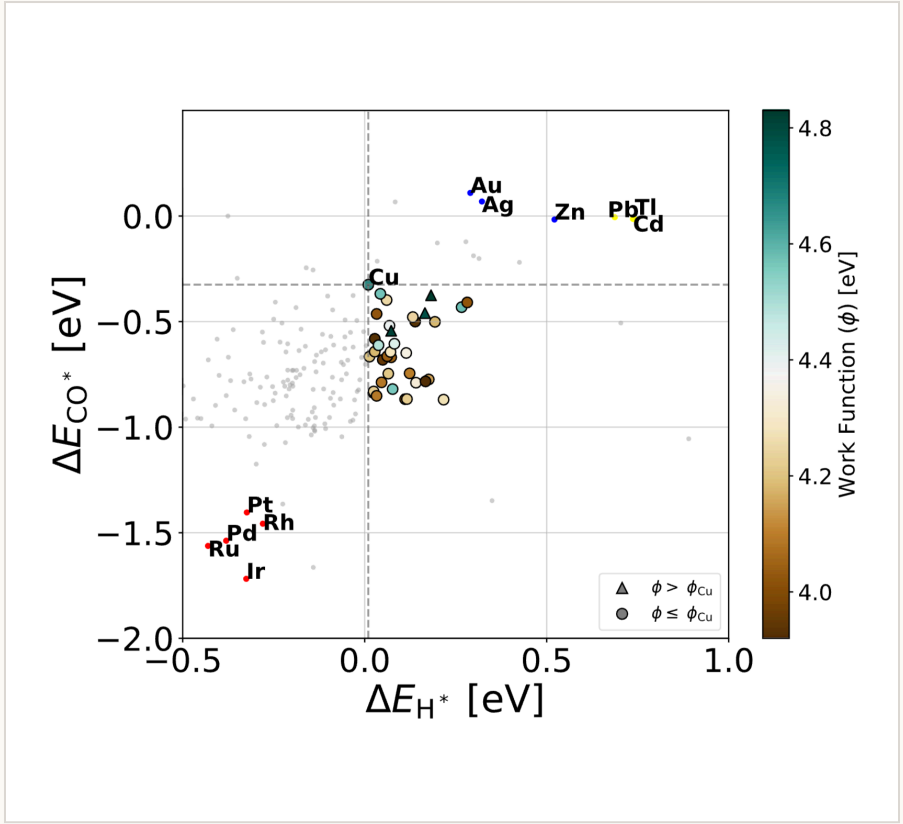


Fig. 4 · DFT binding energies; colour = work function φ

RESULTS

Computational cost

~600

single-point DFT calculations in total

14k

core-hours (~1 h per job on 24 cores)

10^6

calculations for a full high-throughput DFT workflow

RESULTS

Top ten candidates

#	Formula	MP ID	ΔE_{H*}	ΔE_{CO*}	φ [eV]	Score
1	**ZrCuSiAs**	mp-9929	0.17	−0.78	3.92	**1.39**
2	**$MgCu_5$**	mp-1185860	0.14	−0.50	3.93	**1.20**
3	**In_3Pd_5**	mp-22146	0.11	−0.87	4.04	**1.18**
4	**$Y(AlCu)_6$**	mp-1216210	0.05	−0.68	3.95	**1.16**
5	**CdPd**	mp-1696	0.03	−0.58	3.93	**1.11**
6	**$ZnAgPd_2$**	mp-1187926	0.18	−0.77	4.15	**1.04**
7	**SnPd**	mp-2369	0.06	−0.66	4.04	**1.02**
8	**$Zn_{35}Cu_{17}$**	mp-1216020	0.07	−0.67	4.05	**1.02**
9	**In_7Pd_3**	mp-568655	0.03	−0.85	4.09	**1.01**
10	**Cu_6Sn_5**	mp-1233	0.05	−0.79	4.09	**0.99**
ref	**Cu**	mp-30	0.01	−0.33	4.67	–

RANKING SCORE

$$0.5 \cdot (1 - \Delta E_{CO*}) + \Delta E_{H*} + (1 - \varphi)$$

Min–max normalised descriptors; CO* weighted by ½ since only moderate binding is needed

All ten are Cu- or Pd-containing alloys

Table 1 · binding energies in eV · Karlsson et al. 2026

Limitations of the approach

Stability

- Pourbaix stability is necessary but not sufficient
- Kinetic stability and metastable phases not considered
- Inherits errors in Materials Project reference energies

Two-descriptor scheme

- Top ten are strong/weak-binder alloys — a strategy already explored without beating Cu
- Only the most stable single-adsorbate site is captured
- Misses site effects such as CO–CO distance on adjacent Cu atoms

DISCUSSION

An open design space

Many stable materials share copper’s descriptor profile — yet none has clearly outperformed it.

Either

the descriptors miss key physics of CO_2RR

Or

experiments have not yet targeted the right materials

CONCLUSIONS

Summary

01 Three physically motivated filters narrow ~150,000 materials to ten candidates

02 Open databases and a pre-trained MLIP reduce DFT to ~600 single-point calculations

03 The design space is vast; new descriptors are needed, and this workflow can test them

QUESTIONS

Thank you

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